

In this lecture we solve linear systems using Gaussian elimination by reducing them to a simpler form (row echelon form) and back-substitution (if the linear system is consistent) to find their solution set. We begin with definitions needed and solve three different examples of linear system. Although we give more details in this lecture notes, in the future, we shall assume you are fluent with Gaussian elimination and leave all the details for you to complete.

Gaussian elimination

Definition 18. An element in an augmented matrix associated to a linear system is called a pivot (of a row) if it is the first non-zero element in the row.

Examples Below we have identified (by putting a square around them) the pivots in each row of our given augmented matrices.

$$\left(\begin{array}{cccc|c} \boxed{2} & 8 & 3 & 0 & 7 \\ 0 & 0 & \boxed{1} & 8 & 5 \\ 0 & 0 & 0 & \boxed{3} & -4 \end{array}\right), \left(\begin{array}{cccc|c} 0 & \boxed{5} & 2 & 2 & 0 \\ 0 & 0 & 0 & 0 & \boxed{-10} \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & \boxed{7} & 0 & 5 \end{array}\right), \left(\begin{array}{ccc|c} \boxed{4} & 1 & 0 & 5 \\ 0 & 0 & \boxed{1} & 0 \\ 0 & 0 & 0 & 0 \end{array}\right), \left(\begin{array}{cccc|c} 0 & 0 & 0 & \boxed{1} & 4 \\ \boxed{9} & 0 & 0 & 0 & 1 \end{array}\right)$$

Definition 19. An augmented matrix is in row echelon form if it satisfies the following two conditions:

- All rows (if any) that have no pivots or contain only zeros are grouped at the bottom of the augmented matrix;
- Each pivot appears to the right of any pivot in a row above it (if any).

Examples In the list of augmented matrices above, the first and the third are in row echelon form. The second and fourth augmented matrices are not in row echelon form. The second augmented matrix does not satisfy both conditions, while the fourth augmented matrix does not satisfy the second condition.

Solving linear system (1) Given linear system below.

$$\begin{aligned} x_3 + 2x_4 &= -1 \\ 2x_1 + 2x_2 + 7x_3 + 4x_4 &= 3 \\ x_1 + x_2 + 5x_3 + 5x_4 &= 0 \end{aligned}$$

its augmented matrix is

$$\left(\begin{array}{cccc|c} 0 & 0 & 1 & 2 & -1 \\ 2 & 2 & 7 & 4 & 3 \\ 1 & 1 & 5 & 5 & 0 \end{array}\right)$$

Let us denote by R_1 , R_2 and R_3 the first row, second row and third row, respectively.

1. The left-most non-zero column on the above augmented matrix is the first column and the first pivot in this column is 2 in the second row and so we shall interchange row 1 and 2 ($R_1 \leftrightarrow R_2$).

$$\left(\begin{array}{cccc|c} 2 & 2 & 7 & 4 & 3 \\ 0 & 0 & 1 & 2 & -1 \\ 1 & 1 & 5 & 5 & 0 \end{array}\right)$$

2. We shall replace R_3 by $2R_3 - R_1$ ($R_3 \rightarrow 2R_3 - R_1$).

$$\left(\begin{array}{cccc|c} 2 & 2 & 7 & 4 & 3 \\ 0 & 0 & 1 & 2 & -1 \\ 0 & 0 & 3 & 6 & -3 \end{array} \right)$$

3. We shall replace R_3 by $R_3 - 3R_2$ ($R_3 \rightarrow R_3 - 3R_2$).

$$\left(\begin{array}{cccc|c} \boxed{2} & 2 & 7 & 4 & 3 \\ 0 & 0 & \boxed{1} & 2 & -1 \\ 0 & 0 & 0 & 0 & 0 \end{array} \right)$$

The above augmented matrix is in row echelon form and only columns 1 and 3 have pivots, their associated variables x_1 and x_3 are said to be dependent while variables x_2 and x_4 whose associated columns have no pivots are called free variables.

We now write the linear system corresponding to the above augmented matrix.

$$\begin{aligned} 2x_1 + 2x_2 + 7x_3 + 4x_4 &= 3 \\ x_3 + 2x_4 &= -1 \\ 0 &= 0 \end{aligned}$$

‘Moving’ all free variables to the right we get:

$$\begin{aligned} 2x_1 + 7x_3 &= 3 - 2x_2 - 4x_4 \\ x_3 &= -1 - 2x_4 \end{aligned}$$

Let $x_2 = \lambda$ and $x_4 = \mu$ with $\lambda, \mu \in \mathbb{R}$.

We then have $x_3 = -1 - 2\mu$ and $x_1 = \frac{1}{2}(3 - 2\lambda - 4\mu - 7(-1 - 2\mu)) = 5 - \lambda + 5\mu$. We thus have

infinitely many solutions. The solution can be written as $\begin{pmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{pmatrix} = \begin{pmatrix} 5 \\ 0 \\ -1 \\ 0 \end{pmatrix} + \lambda \begin{pmatrix} -1 \\ 1 \\ 0 \\ 0 \end{pmatrix} + \mu \begin{pmatrix} 5 \\ 0 \\ -2 \\ 1 \end{pmatrix}$

(where $\lambda, \mu \in \mathbb{R}$) in vector form and so the solution set is:

$$\left\{ \begin{pmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{pmatrix} = \begin{pmatrix} 5 \\ 0 \\ -1 \\ 0 \end{pmatrix} + \lambda \begin{pmatrix} -1 \\ 1 \\ 0 \\ 0 \end{pmatrix} + \mu \begin{pmatrix} 5 \\ 0 \\ -2 \\ 1 \end{pmatrix} \mid \lambda, \mu \in \mathbb{R} \right\}$$

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Solving linear system (2) Given linear system below.

$$\begin{aligned} x + 2y + z &= 8 \\ -x + 3y - 2z &= 1 \\ 3x + 4y - 7z &= 10 \end{aligned}$$

We begin with its augmented matrix and reduce it to row echelon form

$$\begin{aligned} \left(\begin{array}{ccc|c} 1 & 2 & 1 & 8 \\ -1 & 3 & -2 & 1 \\ 3 & 4 & -7 & 10 \end{array} \right) &\xrightarrow{R_2 \rightarrow R_2 + R_1} \left(\begin{array}{ccc|c} 1 & 2 & 1 & 8 \\ 0 & 5 & -1 & 9 \\ 3 & 4 & -7 & 10 \end{array} \right) \\ &\xrightarrow{R_3 \rightarrow R_3 - 3R_1} \left(\begin{array}{ccc|c} 1 & 2 & 1 & 8 \\ 0 & 5 & -1 & 9 \\ 0 & -2 & -10 & -14 \end{array} \right) \\ &\xrightarrow{R_3 \rightarrow 5R_3 + 2R_2} \left(\begin{array}{ccc|c} 1 & 2 & 1 & 8 \\ 0 & 5 & -1 & 9 \\ 0 & 0 & -52 & -52 \end{array} \right) \end{aligned}$$

The last augmented matrix is in triangular form, in particular row echelon form and its associated linear system is:

$$\begin{aligned} x + 2y + z &= 8 \\ 5y - z &= 9 \\ -52z &= -52 \end{aligned}$$

Solve it using back-substitution and get $x = 3, y = 2$ and $z = 1$, thus the solution set is $\{(3, 2, 1)\}$.

Solving linear system (3) Given linear system below.

$$\begin{aligned} 2y - z &= 1 \\ 4x + y - z &= 4 \\ 4x + 5y - 3z &= 7 \end{aligned}$$

We begin with its augmented matrix and reduce it to row echelon form

$$\begin{aligned} \left(\begin{array}{ccc|c} 0 & 2 & -1 & 1 \\ 4 & 1 & -1 & 4 \\ 4 & 5 & -3 & 7 \end{array} \right) &\xrightarrow{R_1 \leftrightarrow R_2} \left(\begin{array}{ccc|c} 4 & 1 & -1 & 4 \\ 0 & 2 & -1 & 1 \\ 4 & 5 & -3 & 7 \end{array} \right) \\ &\xrightarrow{R_3 \rightarrow R_3 - R_1} \left(\begin{array}{ccc|c} 4 & 1 & -1 & 4 \\ 0 & 2 & -1 & 1 \\ 0 & 4 & -2 & 3 \end{array} \right) \\ &\xrightarrow{R_3 \rightarrow R_3 - 2R_2} \left(\begin{array}{ccc|c} 4 & 1 & -1 & 4 \\ 0 & 2 & -1 & 1 \\ 0 & 0 & 0 & 1 \end{array} \right) \end{aligned}$$

The last augmented matrix is in row echelon form, and the last row would imply we have equation $0 = 1$, and therefore our linear system is inconsistent and so its solution set is $\emptyset$.

Below is the Gaussian elimination algorithm that can be used to reduce augmented matrices of linear systems into row echelon form.

Gaussian elimination algorithm

Step 1: Locate the leftmost non-zero column and the first pivot of that column.

Step 2: If necessary (that is, if the pivot is not in the first row), interchange the first row with the row that contains that pivot.

Step 3: Make all other entries in the leftmost non-zero column zero by using multiples of row 1.

Step 4: Repeat steps 1 to 3 ignoring the top rows whose pivots (if any) have zeros below them.